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## Smart Contracts Part 1: Introduction to Solidity

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The two exercises below are designed to give you basic practical experience with the Solidity language. They are not part of an assignment and your solutions do not need to be uploaded for grading. Despite this, completion of these exercises is strongly encouraged, as it will give you a solid base for the development of the course project. It is recommended to use [Remix](#).

The [introduction to smart contracts](#) in the Solidity documentation is a good starting point.

### Exercise 1 -

Write a simple contract that manages a crowdfunding campaign. The contract should fulfil the following requirements:

- Receive pledges containing Ether from users
- Keep track of who donated how much
- Set a campaign goal
- End the campaign at some point

Once you have your contract working, some interesting points to think about are:

- How do we identify the owner of the contract, i.e., the campaign manager?
- Can donors execute all functions in the contract?
- How can we return the list of donors to the owner?

### Exercise 2 -

Write a simple contract that oversees a crowdfunding campaign. The contract should fulfil the following requirements:

- Run a crowdfunding campaign
- Log each individual donor via an event

Events allow us to have off-chain communication, such that we can communicate with external applications. You can use the contract that you wrote in the previous exercise to do the tasks related to running the campaign itself.